

Soil Health Card Scheme (SHC) – Detailed Government Scheme Report

Scheme Name

Soil Health Card Scheme (SHC)

Launched By: Government of India

Ministry: Ministry of Agriculture and Farmers Welfare

Department: Department of Agriculture and Farmers Welfare

Official Portal:

[Soil Health Card Portal](#)

Government Information:

[National Informatics Centre – Soil Health Card Portal](#)

1. Introduction

The Soil Health Card Scheme (SHC) is a flagship agricultural initiative launched by the Government of India on 19 February 2015. The scheme aims to provide farmers with detailed information about the nutrient status and quality of their soil. It helps farmers understand the fertility of their land and recommends appropriate fertilizers and soil nutrients for improving crop productivity.

The Soil Health Card acts like a health report for agricultural land. Just as doctors analyze human health and provide treatment suggestions, soil experts test soil samples and provide recommendations to improve soil fertility and crop production.

2. Background of the Scheme

Agriculture is highly dependent on soil quality. Many farmers traditionally use fertilizers without knowing the actual nutrient requirements of the soil. Excessive use of fertilizers leads to:

- Soil degradation
- Nutrient imbalance
- Reduced crop yield
- Environmental pollution
- Increased farming costs

To solve these problems, the Government introduced the Soil Health Card Scheme to encourage scientific farming practices and balanced nutrient management.

3. Objectives of the Soil Health Card Scheme

The major objectives of the scheme are:

1. Improve Soil Fertility

To identify nutrient deficiencies and improve soil health.

2. Promote Balanced Fertilizer Use

To encourage scientific and need-based fertilizer application.

3. Increase Agricultural Productivity

To help farmers achieve higher crop yields.

4. Reduce Cultivation Cost

To minimize unnecessary expenditure on fertilizers.

5. Support Sustainable Agriculture

To maintain soil quality for future generations.

6. Strengthen Soil Testing Infrastructure

To improve laboratory facilities and testing services.

7. Create Awareness Among Farmers

To educate farmers about soil management and nutrient requirements.

4. Launch Details

Particulars	Details
Scheme Name	Soil Health Card Scheme
Launch Date	19 February 2015
Launched By	Government of India
Ministry	Ministry of Agriculture and Farmers Welfare
Beneficiaries	Farmers
Coverage	Entire India
Card Validity	3 Years
Mode	Soil Testing and Card Distribution

5. What is a Soil Health Card?

A Soil Health Card is an official document issued to farmers that contains detailed information about the condition of their soil.

The card provides:

- Soil nutrient status
- Fertility level
- Soil pH value
- Organic carbon content
- Fertilizer recommendations
- Crop-wise nutrient suggestions
- Soil improvement measures

It helps farmers make informed decisions regarding crop selection and fertilizer usage.

6. Information Available in Soil Health Card

The Soil Health Card contains information on approximately 12 soil parameters:

Macronutrients

1. Nitrogen (N)
2. Phosphorus (P)
3. Potassium (K)

Secondary Nutrients

4. Sulphur (S)

Micronutrients

5. Zinc (Zn)
6. Iron (Fe)
7. Copper (Cu)
8. Manganese (Mn)
9. Boron (B)

Physical Parameters

10. pH Level
11. Electrical Conductivity (EC)
12. Organic Carbon (OC)

These parameters help determine soil fertility and nutrient availability.

7. Eligibility Criteria

The following farmers are eligible:

Eligible Beneficiaries

- **Small farmers**
- **Marginal farmers**
- **Medium farmers**
- **Large farmers**
- **Individual landowners**
- **Farmers across all states and union territories**

Basic Requirement

- **Must own or cultivate agricultural land.**
- **Must submit soil samples for testing.**

The scheme is available to all farmers across India.

8. Features of the Scheme

Nationwide Coverage

The scheme covers all states and union territories.

Scientific Soil Testing

Samples are tested in authorized laboratories.

Free Soil Analysis

Farmers receive scientific reports and recommendations.

Multilingual Cards

Cards are available in multiple Indian languages.

Digital Platform

Farmers can access information online.

Regular Updates

A new card is generally issued every three years.

Mobile Soil Testing Laboratories

Testing services are available even in remote areas.

9. Working Process of the Scheme

Step 1: Soil Sample Collection

Soil samples are collected from farmers' fields.

Collection Standards

Area Type Grid Size

Irrigated Land 2.5 Hectares

Rainfed Land 10 Hectares

Samples are collected using GPS-based methods.

Step 2: Laboratory Testing

The samples are sent to:

- **Soil Testing Laboratories**
- **Mobile Soil Testing Labs**
- **Agricultural Research Centers**

Experts analyze the nutrient content of soil.

Step 3: Soil Analysis

Laboratories determine:

- **Nutrient deficiencies**
- **Soil fertility level**
- **Soil health condition**
- **Required nutrients**

Experts prepare recommendations based on scientific analysis.

Step 4: Soil Health Card Generation

The card is generated through the Soil Health Card Portal and includes:

- **Farmer details**
 - **Soil nutrient status**
 - **Recommended fertilizers**
 - **Crop suggestions**
 - **Soil improvement measures**
-

Step 5: Distribution to Farmers

The completed Soil Health Card is distributed to farmers through:

- Agriculture Department
 - Village Offices
 - Common Service Centers (CSCs)
 - Online Portal
-

10. Benefits of the Soil Health Card Scheme

Agricultural Benefits

Better Crop Selection

Farmers can choose crops suitable for their soil.

Higher Productivity

Improved nutrient management increases yield.

Balanced Fertilizer Usage

Reduces overuse of chemical fertilizers.

Improved Soil Fertility

Maintains long-term soil health.

Scientific Farming

Encourages modern agricultural practices.

Economic Benefits

Reduced Fertilizer Cost

Farmers purchase only necessary nutrients.

Increased Income

Higher productivity leads to better earnings.

Reduced Crop Losses

Proper nutrient management improves crop health.

Efficient Resource Utilization

Minimizes waste of agricultural inputs.

Environmental Benefits

Reduced Soil Pollution

Prevents excessive fertilizer use.

Better Water Quality

Reduces nutrient runoff into water bodies.

Sustainable Agriculture

Supports long-term environmental conservation.

Balanced Nutrient Management

Protects natural soil properties.

11. Advantages of Soil Testing

Soil testing helps farmers:

- **Identify nutrient deficiencies**
- **Improve fertilizer efficiency**
- **Enhance crop growth**
- **Increase agricultural output**
- **Reduce production costs**
- **Maintain soil sustainability**

Experts estimate that balanced fertilizer use can significantly improve productivity and reduce input costs.

12. Role of Technology in the Scheme

The Soil Health Card Portal supports:

- **Online registration**
- **Sample tracking**
- **Soil card generation**
- **Multilingual reports**
- **Digital soil maps**
- **Fertilizer recommendations**
- **Dashboard and MIS reports**

The portal is available through web and mobile applications.

13. Challenges of the Scheme

Despite its benefits, some challenges exist:

Limited Awareness

Many farmers are unaware of soil testing benefits.

Testing Infrastructure Issues

Some regions have insufficient testing laboratories.

Delays in Report Generation

Large sample volumes may delay results.

Adoption Problems

Farmers may not fully follow recommendations.

Incomplete Soil Indicators

Some important biological and physical soil properties are not included.

Technical Constraints

Data collection and monitoring can be challenging in remote areas.

14. Government Measures for Improvement

The Government has introduced:

- **Mobile Soil Testing Laboratories**
- **Digital Soil Health Card Portal**
- **Farmer Awareness Programs**
- **Laboratory Capacity Expansion**
- **Integration with Agricultural Databases**
- **GPS-Based Soil Sampling**

These measures improve accessibility and efficiency.

15. Impact of the Scheme

The Soil Health Card Scheme has contributed to:

- **Better fertilizer management**
- **Increased crop productivity**
- **Reduced cultivation expenses**
- **Improved soil fertility awareness**
- **Promotion of sustainable agriculture**

- Scientific decision-making among farmers

The Government initially planned coverage for nearly 14 crore farmers across India.

16. Importance of Soil Health Card in Modern Agriculture

The Soil Health Card plays an important role in:

- Precision farming
- Sustainable agriculture
- Organic farming support
- Resource conservation
- Climate-resilient agriculture
- Nutrient management planning

It serves as a scientific guide for farmers to maintain healthy soil and achieve higher agricultural productivity.

17. Conclusion

The Soil Health Card Scheme is one of India's most important agricultural initiatives aimed at improving soil fertility and promoting sustainable farming practices. By providing scientific information about soil nutrient status and fertilizer requirements, the scheme helps farmers increase productivity, reduce cultivation costs, and protect environmental resources. The program encourages balanced fertilizer use, enhances soil health, and supports long-term agricultural development. Through digital technology, laboratory testing, and farmer awareness programs, the Soil Health Card Scheme contributes significantly to modernizing Indian agriculture and ensuring food security.

Quick Summary

Particular	Details
Scheme Name	Soil Health Card Scheme
Launch Year	2015
Ministry	Ministry of Agriculture and Farmers Welfare
Beneficiaries	Farmers
Purpose	Soil Testing and Nutrient Management
Coverage	Entire India

Particular	Details
Validity	3 Years
Main Benefit	Scientific Fertilizer Recommendation
Target	Improved Soil Fertility and Crop Productivity
Portal	Soil Health Card Portal